Aakash Patil

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Education

Masters of Science in Computer Science Aug 2025 – Present
George Mason University, Fairfax VA *GPA: 3.8*

- **Relevant Coursework:** Analysis of Algorithms, Machine Learning, Artificial Intelligence, Component-Based Software Development, Computer Systems and Systems Programming

Bachelors in Computer Engineering (Minor in AI&ML) Aug 2021 – May 2025
Xavier Institute of Engineering, Mumbai *GPA: 3.2*

- **Relevant Coursework:** Data Structures, Database Management Systems, Statistics & Probability, Big Data Analysis, Machine Learning, Operating Systems, Computer Networks

Technical Skills

Languages: Python, SQL, C++
Data Tools: Pandas, NumPy, SciPy, Scikit-learn, Excel
Cloud Platforms: AWS, Azure, Google Cloud Platform
Databases: MySQL, MongoDB, DynamoDB, AWS S3, NoSQL
Platforms & Other: Docker, Jenkins, FastAPI, Streamlit, AWS Lambda, Git, ETL Pipelines **Analytics & Visualization**: Tableau, Power BI, Matplotlib, Seaborn

Experience

Data Analytics Intern June 2023 – July 2023
IBM SkillsBuild *Mumbai*

- Conducted exploratory data analysis on image datasets using Pandas and Matplotlib, identifying quality metrics that guided preprocessing decisions
- Built data visualization dashboards to track model training performance, communicating insights to cross-functional team members
- Cleaned and transformed 10K+ image records through standardized ETL workflows, improving data consistency for downstream pipelines

Projects

StyleSnap – AI Outfit Detection & Recommendation | *Python, Polars, FastAPI, Gemini API* Feb 2025

- Processed and structured fashion image metadata using Polars and NumPy, building data pipelines to feed a YOLOv8 detection model
- Integrated Google Gemini API to generate context-aware outfit recommendations, transforming raw detection outputs into user-facing insights
- Deployed a FastAPI-based application enabling users to upload images and receive ranked recommendations, tracking usage metrics via logging

PPXAI – Privacy-Preserving Explainable AI | *Python, Pandas, SHAP, LIME* July 2024 – May 2025

- Performed statistical analysis on healthcare prediction models, evaluating accuracy, precision, and recall across patient subgroups
- Generated SHAP-based feature importance reports to identify the top contributing variables in clinical prediction models, enabling stakeholder review
- Documented analytical findings on privacy-accuracy trade-offs, producing reproducible analysis notebooks for the research team

ERS – Content-Based Recommendation System | *Python, NumPy, Pandas, Scikit-learn* Jan 2023 – May 2023

- Engineered TF-IDF feature vectors from unstructured text data and computed cosine similarity matrices to power a recommendation engine
- Performed A/B-style evaluation of recommendation quality metrics, iterating on feature selection to improve user satisfaction

Image Super Resolution Analytics | *Python, Pandas, Matplotlib, TensorFlow* July 2023 – Nov 2023

- Tracked and visualized GAN training metrics (PSNR, SSIM, loss curves) across 50+ training epochs using Matplotlib dashboards
- Built a Streamlit reporting tool for non-technical users to compare input vs. output image quality side-by-side